



Making Solar Panels Thrive with AI-Powered Multispectral Magic 🌈

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Motivation

- Our previous project EAGLE proves that AI-driven multispectral imaging can spot PV module defects with superhero-level accuracy. But why stop there? Let's make it an ecosystem that boosts operations, finances, and planet-friendliness, all while keeping things sunny.
- EAGLE turns defect detection from a nerdy diagnostic tool into a cheerful engine of sustainability, financial trust, and PV longevity. Solar panels deserve happiness too.

Digital Twins (because real twins are expensive ;-)

- Real-time health tracking of PV systems
- Lifetime yield prediction (crystal ball not required)
- Insurance risk assessment

Finance & Risk (the fun part, surprisingly):

- Transparent, image-based diagnostics
- Claim validation that makes insurers say 'Finally!'
- Data-driven risk modelling

Circular Economy Awesomeness:

- Spot which modules can enjoy a second life
- Help power rural areas and off-grid systems
- Guide recyclers by identifying degradation patterns

Multi-Spectral Image Capturing

Visual Image Setup:

- UVf image confirms there is only one cell crack (oxygen bleaching on the cell crack area)

Electroluminescence Image:

- A power supply connected to PV module, IR emissions from the cells
- A cell crack and damaged metallization (at the rear side of the cell) are visible in the EL image

Ultraviolet Fluorescence Image Setup:

- UVf image confirms there is only one cell crack (oxygen bleaching on the cell crack area)
- When UV light (high energy) is absorbed by degradation by products of polymeric materials, which may have fluorescence effect, they emit photons in the visible range (lower energy)

Example:

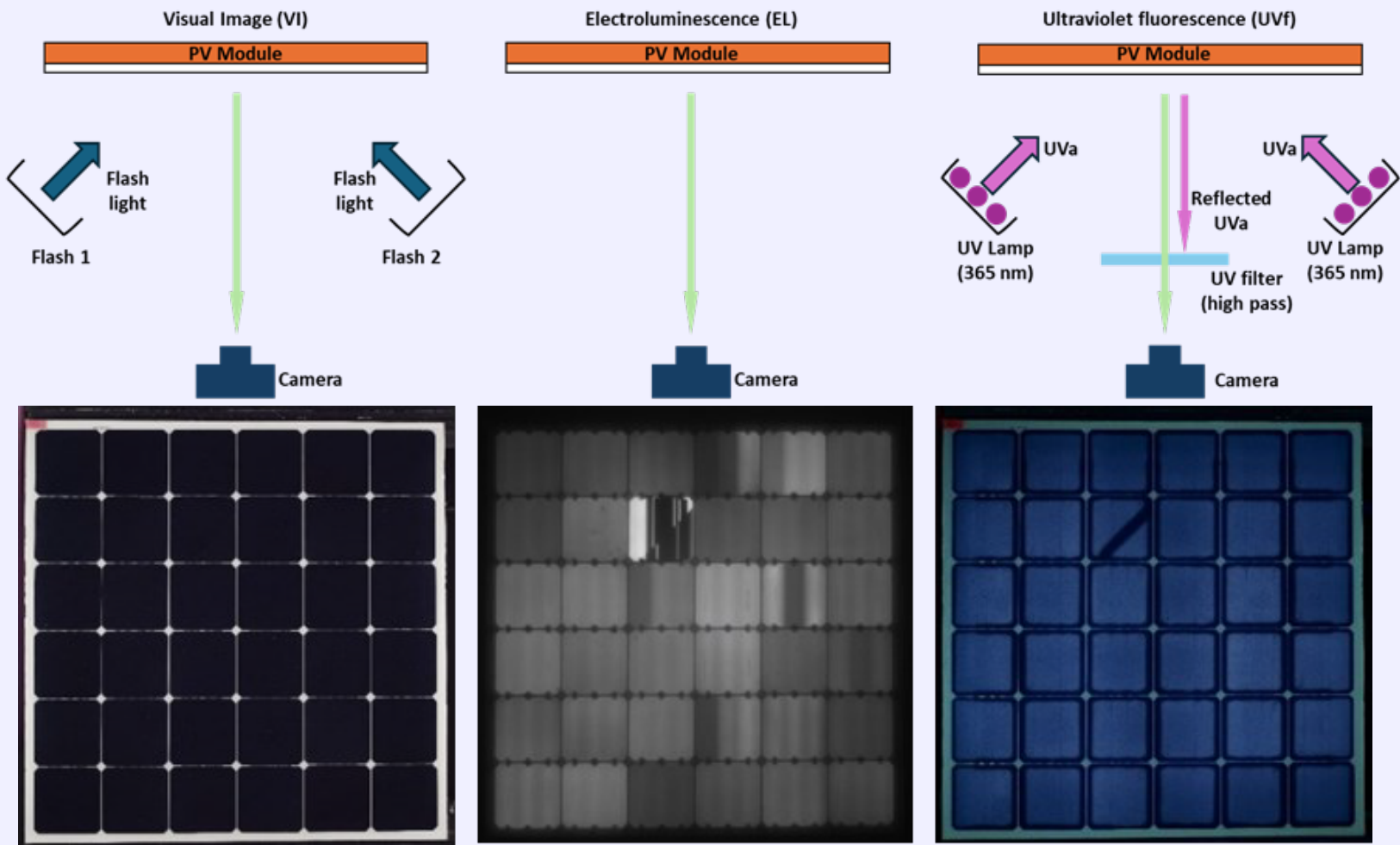
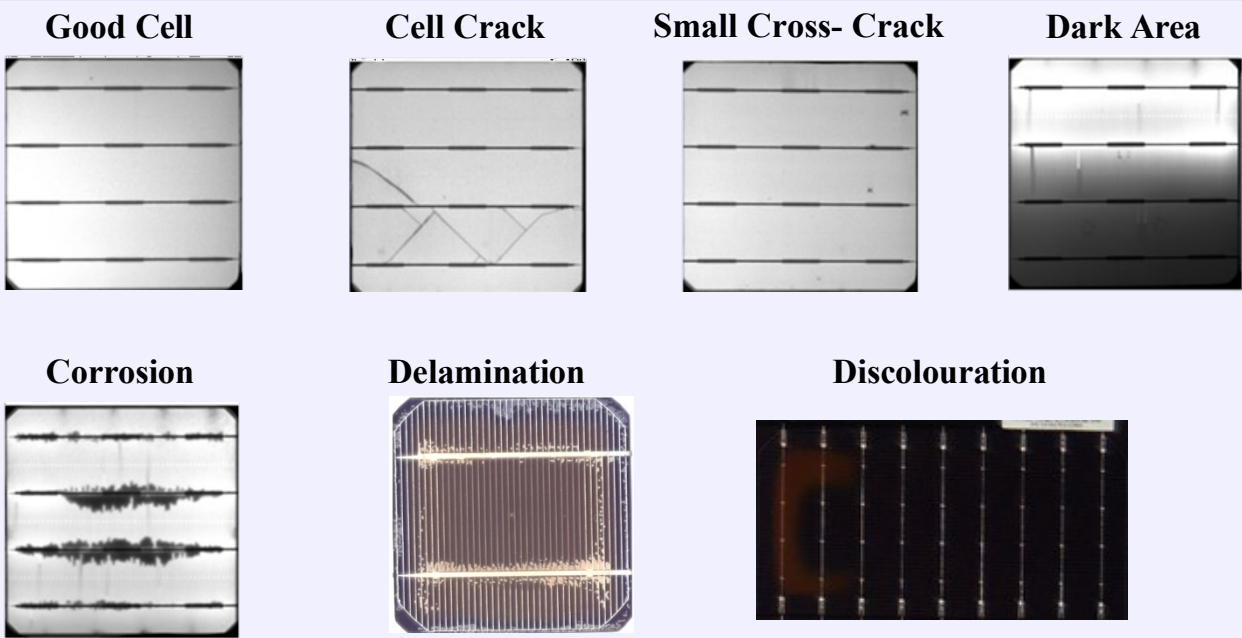


Figure 1. (top) Schematic view of visual inspection (VI), electroluminescence (EL) and ultraviolet fluorescence (UVf) setups. (bottom) VI, EL, and UVf images of an IBC module after a long-term outdoor exposure.

- Using multispectral image like in this case gives possibility to separate damaged metallization and cell crack without any hesitation.

Cell Classification



- Discoloration can be identified through both VI and UVf images, depending on specific conditions such as encapsulant type. In cases where discoloration is intense, it may also lead to oxidation of the front metallization, which can be observed in the EL image (red frames in figure below). In such instances, cells are labelled with multiple failure modes.

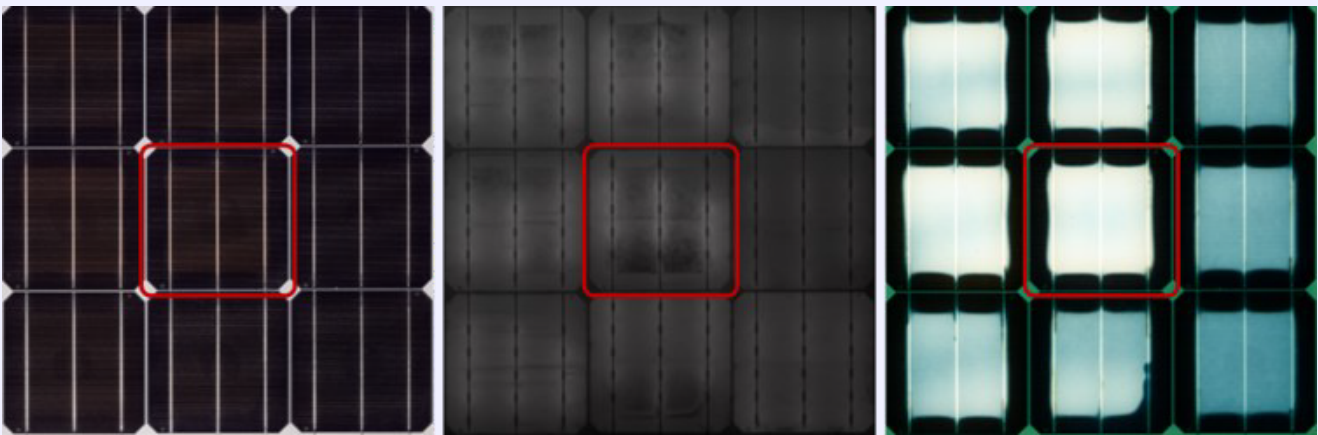


Figure 2. (left) VI, (middle) EL, and (right) UVf images of a section of a module after 10 years of operation under frequent shading. The cell with the red frame is labeled as "discolouration" and "corrosion".

Performance

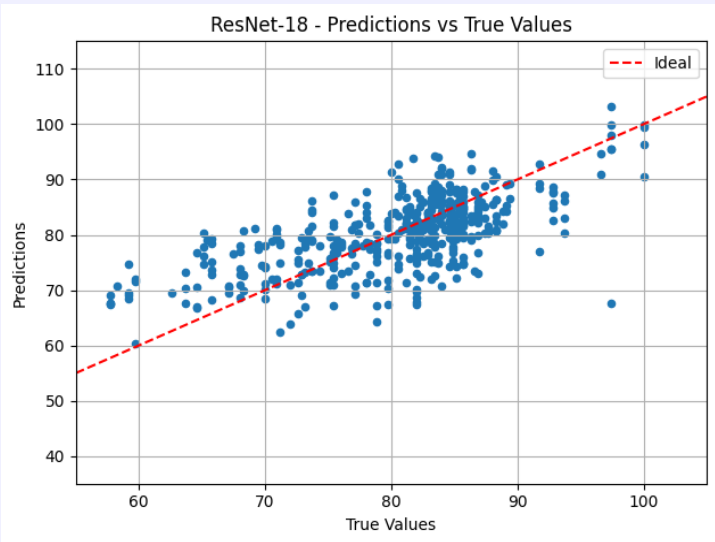


Figure 2. Max Power prediction based on EL, VI and UVf images. Predicted vs true Max Power values, with our models we can predict with +/- 4.3% accuracy the performance of the a panel based solely on an image

Results

- The classification model, **ChannelViT**, was trained on the Duramat dataset [1] and tested on the Infinity dataset [2]. This means the model was trained on different technologies than those used for classification prediction. Despite the model never having encountered images from the Infinity dataset during training, it achieved an impressive prediction accuracy of **94.4%**.

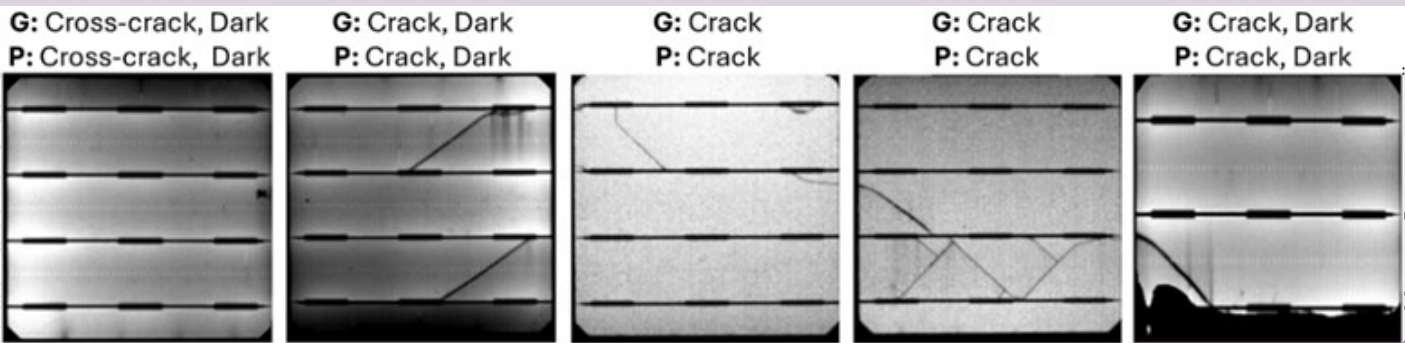


Figure 3. Examples of correct Vision Transformer predictions for the Infinity dataset, where the ground truth (G) is the same as the prediction (P).